



University
of Glasgow

Thursday 16 May 2024
14:00– 16:00 BST
Duration: 2 hours
Timed exam – fixed start time

DEGREE OF MSc

Information Retrieval COMPSCI5011

Answer All Questions

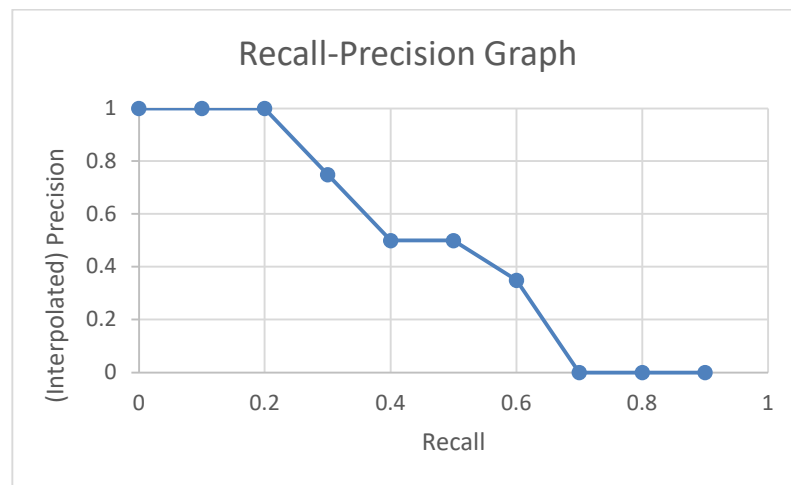
This examination paper is an open book, online assessment and is worth a total of 60 marks.

1.

(a)

- (i) Explain the issue of word variants in text and illustrate it with an example not shown in the lectures. [2]
- (ii) Briefly describe why word variants can affect the performance of an IR system. [1]
- (iii) Explain if the issue mainly affects the precision or recall performance of the system and why. [1]

(b) Consider a query with 10 relevant documents. Consider the following interpolated recall-precision graph evaluating the output of an information retrieval system for that query:



- (i) Provide the precision performance of this IR system for the query after it has retrieved three relevant documents. *Justify your answer.* [1]
- (ii) The user who issued the query starts going through the ranked list of returned documents from the top rank and keeps going down the list. At one point in the ranking, the user realises that all k returned documents by the system are relevant. What is the k value? *Justify your answer.* [2]
- (iii) The IR system has returned 20 documents ranked from rank=1 to rank=20. Indicate each of the rank(s) where the corresponding document at that rank is relevant. Briefly *justify your answer* based on the shown recall-precision graph. [2]

- (iv) Compute the Average Precision (AP) of this IR system for the query. Round to 4 decimal places. *Show your working.*

[1]

- (c) Consider the following corpus of documents:

Doc1: I like Terrier Terrier Terrier

Doc2: I like Terrier and I like the course

Doc3: I like the Terrier course

Doc4: I do not like dogs

Consider that you are using an IR system where stemming is not applied but where the following stopwords are removed: I, and, do, not, the.

Assume that you have constructed an inverted file for all the documents of the corpus. This inverted file should store sufficient information for computing a simple $tf \cdot idf$ term weight, where $w_{ij} = tf_{ij} \cdot \log_2(N/df_i)$.

- (i) Show all the relevant information in the constructed inverted file for the following terms: course, like, terrier.

[3]

- (ii) Give the vector space model representations of all 4 documents in the collection. You need to use $w_{ij} = tf_{ij} \cdot \log_2(N/df_i)$ to weight the terms. Round to 4 decimal places.

Show your workings. Note that $\log_2(1.3333) = 0.4150$, $\log_2(0.75) = -0.4150$, $\log_2(0.5) = -1$ and $\log_2(0.25) = -2$ (**NB:** you may not need all of these).

[4]

- (iii) The user issues the following query to the above IR system:

I like terrier dogs

Using a binary weighting model to weight the query terms (the weight of a term is 1 if present in the query, 0 otherwise), provide the vector space model representation of the above query, clearly showing both the terms and their weights.

NB: You can use any reasonable plain text notation for the vector.

[1]

- (iv) Using your inverted file and the dot product (aka the *inner product*) as a similarity measure, rank the four documents of the corpus by relevance score for the above query. Round to 4 decimal places. *Show your workings.*

[2]

2.

- (a) Assume that Zipf's law holds for a large collection of English documents. Assume that a given term of the collection has a rank of 1021 and a term frequency of 5095 in the collection.

Estimate the probability of occurrence of the two most frequent terms in the collection. *Show your workings, you need not calculate the final values.*

[2]

- (b) Consider a collection with one million tokens (i.e., with 10^6 term occurrences). Suppose the first 64 of these tokens results in a vocabulary size of 40 terms, and the first 2500 tokens results in a vocabulary size of 250 terms.

Use Heap's law with $\beta = 0.5$ to estimate the vocabulary size of the whole collection. *Show your working.* Note that $\sqrt{2500} = 50$.

[3]

- (c) Consider the query "bank interest" and the following term frequencies for the three documents D1, D2 and D3, where the search engine is using raw term frequency (TF) but no IDF:

	bank	interest	jogging	look	raise	rate	river	sailboat
D1	2	2	4	3	0	0	3	0
D2	4	1	3	1	0	0	3	3
D3	6	4	0	0	3	4	0	0

Assume that the system has returned the following ranking: D2, D1, D3. The user judges D1 to be relevant and both D2 and D3 to be non-relevant for this query.

- (i) Show the initial query vector, clearly stating the terms of the vector.
NB: You can write the vector in plain text. For example, a vector of 3 dimensions can be written as:

term₁ term₂ term₃
vector (xx, yy, zz)

[1]

- (ii) Use Rocchio's relevance feedback algorithm (with $\alpha=\beta=\gamma=1$) to provide a revised query vector for "bank interest". Terms in the revised query that have negative weights can be dropped, i.e. their weights can be changed back to 0. Show all your calculations.

[4]

(d) Consider the following three documents.

D1: Rise pay rise pay for employees

D2: I like my pay

D3: Employees like employees

Assume an IR system that only removes stopwords (a, i, for, my). This system also uses the probabilistic Binary Independence Retrieval Model. Recall that in this model, the relevance score of a document to a query is computed using the following RSV formulae:

$$\sum_{i:d_i=1} \log \frac{p_i(1-s_i)}{s_i(1-p_i)}$$

- (i) Using the BIM Model, provide the document-term matrix for D1, D2 and D3.

NB: You can use any reasonable plain text notation for the matrix, clearly labelling the documents and terms.

[2]

- (ii) Assume that the following query Q has been issued by a user to the system:

Q = pay rise

Assume that there is no relevance information available to the system. Calculate the Retrieval Status Value (RSV) of the three documents above for query Q, and rank the documents. Use log base 10.

Show your workings and justify your answer. Round to 4 decimal places. Note that $\log_{10}(0.25)=-0.6020$, $\log_{10}(0.5)=-0.3010$, $\log_{10}(0.75)=-0.1249$, $\log_{10}(2)=0.3010$, $\log_{10}(3)=-0.4771$ (NB: you may not need all of these).

[3]

- (e) Consider the following positional inverted index, where each posting contains docid and term positions.



- (i) Show the bigram paired posting list for the bigram X Y.

[2]

- (ii) For this query, users often prefer documents where query terms occur in a phrase. What is the efficiency advantage of storing a paired posting list for $X\ Y$?

[1]

- (iii) Terrier and other retrieval systems support a query operator $\#uwN$, where N is a distance in terms of tokens. This counts the number of windows of size N in a document where the query tokens both appear. (query tokens can appear in any order).

Show the paired posting list for $\#uw4(X\ Y)$.

[2]

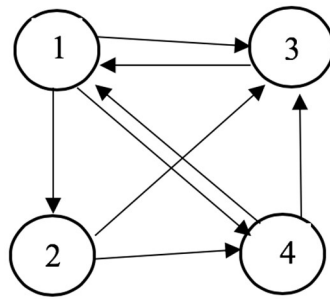
3.

- (a) Provide an example learned model when applying a linear learning to rank approach. You might assume that the model was learnt on a Web collection with 4 features: PL2, Proximity, PageRank and URLlength.

What is the main disadvantage of linear learning to rank models? Justify your answer using your provided example learned model.

[4]

- (b) Consider the following hyperlink structure among 4 web pages:



Consider a query q , which returns all web pages shown in the provided hyperlink structure. Using matrices, explain how you will find the authority scores of each returned web page using the HITS approach. You need to provide *all* required intermediary matrices and give the resulting equations but you do not have to resolve these equations. *Show your workings.*

NB: You can indicate matrices in plain text like a table, or write matrices row by row, e.g. $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ shows a matrix with two columns and two rows where the first row is $[a, b]$ and the second row is $[c, d]$.

[4]

- (c) Consider a neural re-ranking pipeline of the form:

```
first_stage = BatchRetrieve(index, wmodel="BM25", num_results=1000)
reranker = first_stage >> pt.text.get_text(dataset, "text") >> MonoT5()
```

- (i) You run an experiment comparing `first_stage` and `reranker` and obtain the following results from a paired t-test:

	nDCG p-value	Recall@1000 p-value
first_stage	NaN	NaN
reranker	0.001532	NaN

Explain in detail why is the Recall@1000 p-value “NaN” for reranker?

[5]

- (ii) Let us say the Recall@1000 of reranker is 0.431. Provide 3 suggestions on ways that you can improve this value using neural networks. For each suggestion, justify your answer.

[3]

- (d) Contrast lossy and lossless compression of a search engine's inverted index, providing examples of techniques. How might we measure the success of each form of compression?

[4]